PATCH

Your AI coding and security teammate



Release Detective

Naija AI / DevX Arena 2026

The manual investigation takes several steps

90 min

Published median for manual CodeQL alert remediation [1]

Different tasks and timing boundaries. This is industry context, not Patch performance or its human baseline.

01 Understand the route and access policy

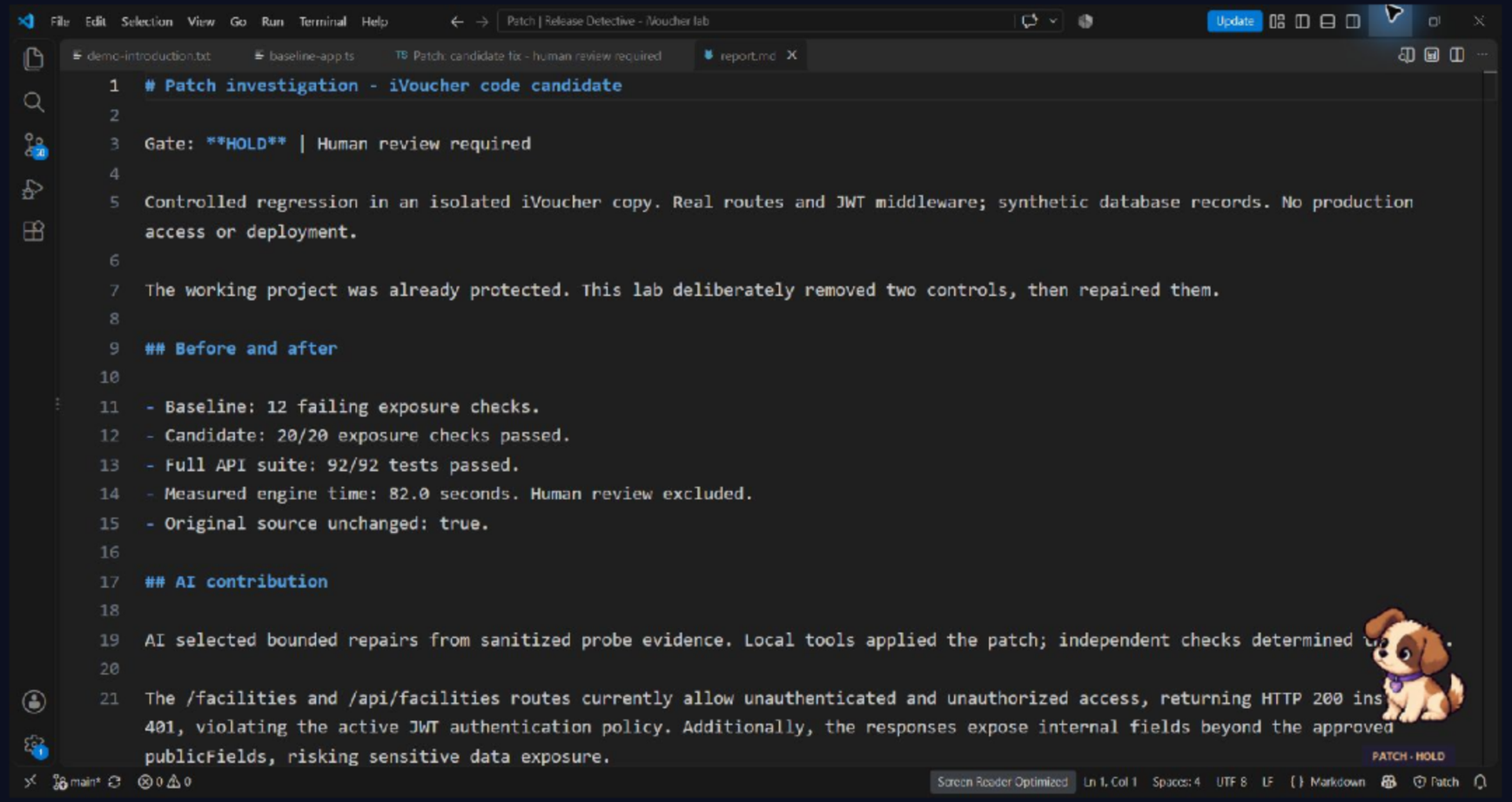
02 Reproduce access and data exposure

03 Prepare the change and check regressions

04 Rerun checks and assemble evidence

OWASP supports the testing workflow [2, 3]; it does not assign minutes to these steps.

Patch keeps evidence and review in the editor



An engineer starts one investigation

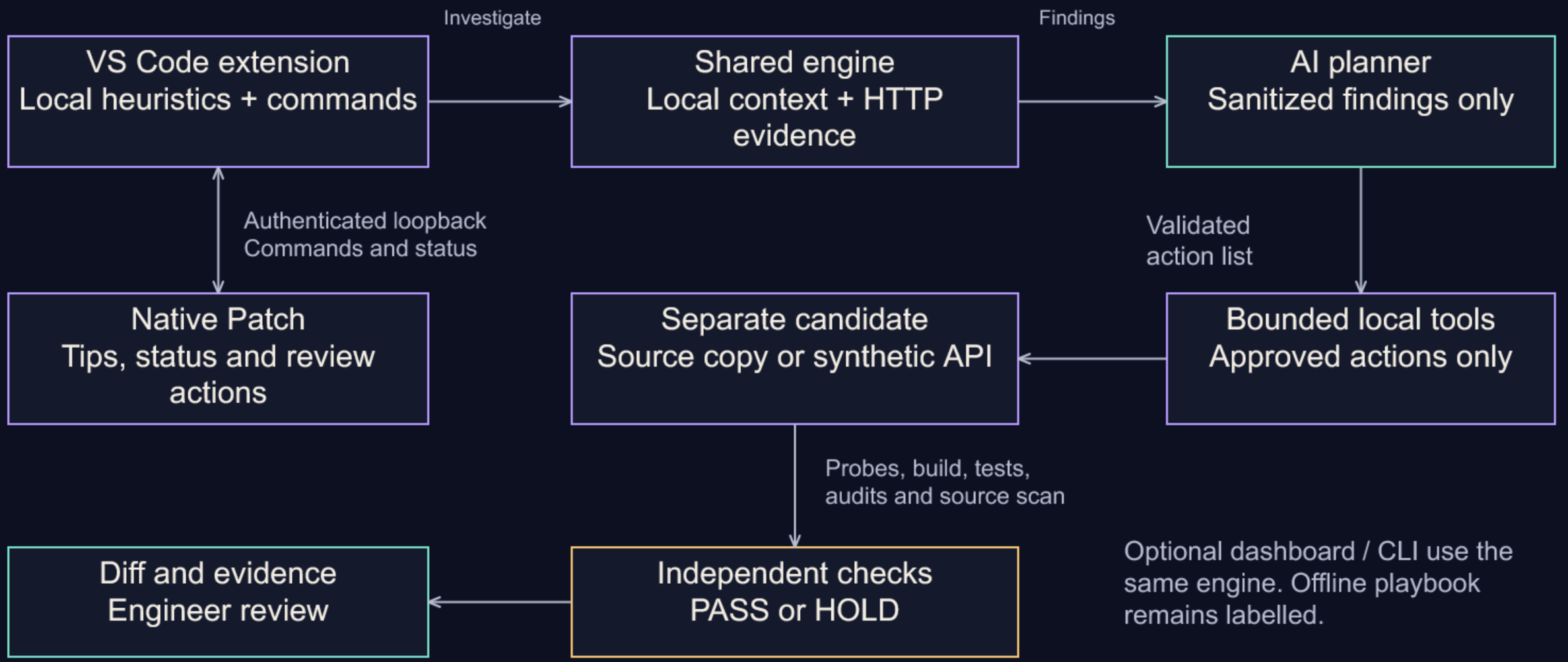
AI selects approved repairs

Tools verify the candidate

The engineer reviews the diff and report

Controlled regression in an isolated copy. Real routes, synthetic records. No deployment.

Patch architecture



API key stays in the extension/backend. No source code, credentials or real records enter model context.

Measured results from two different rehearsals

Workflow	Observed time	Verified outcome
iVoucher source copy	82.0 seconds	92/92 API tests 20 exposure checks included
Smaller policy sandbox	5.188 seconds	36/36 exposure checks Five approved policy repairs

The code candidate builds and passes its tests

The release remains on HOLD: eight production dependency advisories still need attention.

One run per scenario. Human review and dependency remediation are outside these timings.

Potential developer time saved

10–28 min

Estimated saving per smaller sandbox investigation

Includes human review.
A timed manual comparison is still needed.

Component	Time
Manual workflow	15–30 min assumed
Patch automation	5.188 sec measured
Human review	2–5 min assumed
Assisted total	2.09–5.09 min estimated

This estimate is separate from the larger 82-second iVoucher code investigation.

What the prototype taught us

Worked

A real candidate and evidence arrive inside the editor.

The gate keeps dependency blockers visible after a repair.

Next evidence

Time matched manual and assisted tasks, including review and rework.

Validate the recipe in authorized staging with a real database.

A Windows path mismatch and overly broad model actions both surfaced in rehearsal.
We fixed those failures and added regression checks.

Current scope: one facilities recipe, synthetic records, human release authority.

Sources and project evidence

[1] GitHub, 2024

CodeQL alert remediation timing; vendor beta telemetry

[2] OWASP WSTG v4.2

Testing for Bypassing Authorization Schema

[3] OWASP Cheat Sheet Series

Authorization Testing Automation

[4] NIST SP 800-218 v1.1, 2022

Secure Software Development Framework

Full citations, claim boundaries and local measurements are in submission/SOURCES.md and submission/MEASUREMENTS.json in the project repository.